SMART INDIA HACKATHON 2026

TITLE PAGE

- **Problem Statement ID –** 26169
- **Problem Statement Title –** AI-Based Virtual Camera Tracking System for Coarse Alignment of Mobile FSOC Terminals
- **Theme –** Smart Automation
- **PS Category –** Software
- **Team ID –** —
- **Team Name –** ZeroDrift

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IDEA TITLE

- *Proposed Solution (Describe your Idea/Solution/Prototype)*
- *Detailed explanation of the proposed solution*
- *How it addresses the problem*
- *Innovation and uniqueness of the solution*

The eyes and neck of a laser communication terminal

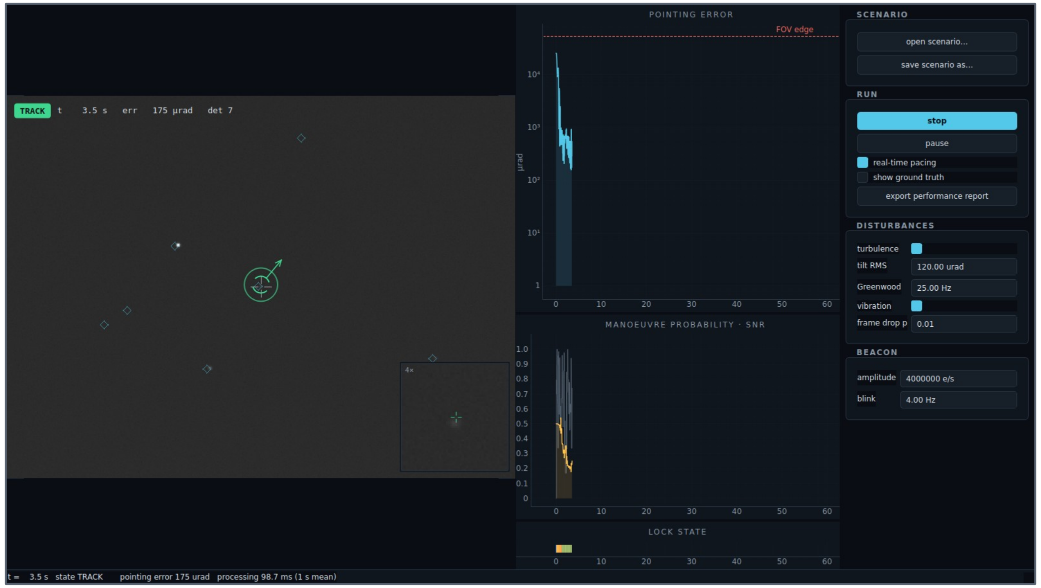
Complete virtual testbed: physics-honest sky — turbulence (Greenwood-correlated), platform vibration, sensor noise, moving beacons on real LEO/UAV paths.

AI-assisted closed-loop tracker: detects the beacon, identifies it, estimates motion, and steers a virtual pan-tilt camera to hold it in the FOV.

Addresses the problem exactly: coarse-alignment algorithms developed and validated with zero optical hardware — ISRO's stated need.

Innovation: beacon identified by its blink signature (hard gate — stars score 0), IMM motion filter, latency-compensating Smith predictor, from-scratch neural track verifier.

Not a concept — already built: GUI console, standalone app, 73 automated tests, full performance reports — everything on this slide is measured, not projected.



Our console, live: locked on the beacon at 175 µrad

1.1 s acquisition time	**98.8 %** lock retention	**0 / 64** wrong-target locks

Median pointing error 207 µrad — beacon kept inside the field of view **100%** of the pass.

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TECHNICAL APPROACH

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- *Technologies to be used (e.g. programming languages, frameworks, hardware)*
- *Methodology and process for implementation (Flow Charts/Images/ working prototype)*

Stack: Python 3.11 · NumPy · OpenCV · PySide6 + pyqtgraph GUI · SGP4 real orbits · from-scratch NumPy neural net · PyInstaller app · pytest (**73 tests**) · GitHub CI

100% open source — zero licensing cost, runs on any laptop

VIRTUAL WORLD sky · stars · clutter turbulence · vibration → **DETECT** top-hat + matched filter + CFAR → **IDENTIFY** blink signature gate + AI verifier → **ESTIMATE** IMM Kalman filter position + velocity → **CONTROL** Smith predictor + feed-forward → **VIRTUAL GIMBAL** slew limits · latency encoder noise

closed loop at the camera's 30 frames per second — real time on a normal laptop

Acquisition state machine

SEARCH → **ACQUIRE** → **TRACK** → **COAST** → **REACQUIRE**

Spiral search over the uncertainty cone → lock → coast through beacon blinks → automatic recovery. Verified: recovers the true beacon even when started 5° off-target among decoys.

Working prototype — tracking a satellite pass

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FEASIBILITY AND VIABILITY

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- *Analysis of the feasibility of the idea*
- *Potential challenges and risks*
- *Strategies for overcoming these challenges*

FEASIBILITY — PROVEN

- Fully working today: 73/73 automated tests pass
- 64-run randomised Monte-Carlo campaign: 100% acquisition, zero wrong locks
- Standalone Windows/Linux app builds in one click
- Runs faster than real time on a normal laptop — no GPU, no cloud, no dataset

CHALLENGES & RISKS

- Heavy turbulence dims and shakes the beacon
- Bright decoys / sun glints can mimic the beacon
- Mount latency destabilises fast tracking loops
- Sharp UAV manoeuvres break simple trackers

OUR STRATEGIES

- Coast-through-fade + automatic re-acquisition spiral
- Blink-signature hard gate: a star scores zero — fooled 0 times in 64 runs
- Smith predictor cancels the measured 40 ms delay
- Dual-model IMM filter adapts to manoeuvres in ~0.3 s

Worst case across all 64 randomised runs: a 0.53×-dim beacon under 1.74× turbulence still acquired in 2.8 s and held 91.7% lock. Every number regenerates from one command — fully reproducible.

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IMPACT AND BENEFITS

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- *Potential impact on the target audience*
- *Benefits of the solution (social, economic, environmental, etc.)*

STRATEGIC

Advances India's next-generation space & defence communications — FSOC pointing algorithms developed sovereignly, without importing optical test benches.

ECONOMIC

A laptop replaces lakhs of rupees of cameras, gimbals and optics — the entire develop–test–validate cycle for FSOC pointing runs in pure software, at zero recurring cost.

RESEARCH & EDUCATION

Any student or ISRO lab can develop, test and compare PAT algorithms instantly — reproducible scenarios, honest ground-truth scoring, open source.

SCALABLE & GREEN

Same engine covers satellite, UAV, ground and maritime links. Pure software — no consumables, no lab energy, no e-waste in the development cycle.

Target audience: ISRO / IN-SPACe optical-communication teams, DRDO, IITs & engineering colleges, and India's growing space-tech startups.

The number that matters: on a tracked ISS pass, coarse stage + modelled fine stage closes the optical link **99.3%** of the time vs **14.7%** without — measured, not asserted.

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RESEARCH AND REFERENCES

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- *Details / Links of the reference and research work*

Research foundations

Kaymak et al., "A Survey on Acquisition, Tracking and Pointing Mechanisms for Mobile FSO," IEEE Comm. Surveys & Tutorials, 2018.

Kaushal & Kaddoum, "Optical Communication in Space: Challenges and Mitigation Techniques," IEEE Comm. Surveys & Tutorials, 2017.

Blackman & Popoli, Design and Analysis of Modern Tracking Systems — IMM filtering, CFAR detection, track management.

Bar-Shalom et al., Estimation with Applications to Tracking and Navigation — Kalman/IMM theory.

Smith, O.J.M., "Closer Control of Loops with Dead Time" — the Smith predictor, 1957.

Vallado & Crawford, "SGP4 Orbit Determination" — AIAA 2008; real ISS TLE propagation.

Andrews & Phillips, Laser Beam Propagation through Random Media — turbulence & scintillation models.

Our work — open and verifiable

Repository: github.com/pranshukesavmishra/SIH-2026

Inside: full source (5,000+ lines, documented) · 73 automated tests · 6 validated scenarios · Monte-Carlo campaign logs · technical report · user manual · demo video.

Reproducibility: every figure in this deck regenerates from one command; every random draw is logged and replayable.

Why ZeroDrift wins on this problem

- ✓ All six mandatory deliverables already exist today
- ✓ Judged numbers, honestly measured against hidden ground truth
- ✓ AI where it earns its place — benchmarked against classical methods